

Power electronics project part b



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Question 1

i.

Calculations

$$Z_n = \frac{R}{1 + j\omega RC}$$

$$\angle Z = -\arctan(\omega RC)$$

For the DC component

$$V_{dc} = 3 \times 100 = 300 \text{ V}$$

For first harmonic

$$Z_1 = \frac{100}{1 + j * 120\pi * 100 * 50 \times 10^{-6}}$$

$$Z_1 = 46.86 \angle -62.05^\circ \Omega$$

$$V_1 = \sqrt{2} V_{1rms} = \sqrt{2} V_{1rms} I_{rms} = \sqrt{2} * \frac{4}{\sqrt{2}} * 46.86 = 187.44$$

$$v_1(t) = 187.44 \cos(2\pi * 60t - 62.05^\circ) = 187.44 \sin(2\pi * 60t - 27.95^\circ) =$$

For second harmonic

$$Z_1 = \frac{100}{1 + j * 240\pi * 100 * 50 \times 10^{-6}}$$

$$Z_1 = 25.64 \angle -75.14^\circ \Omega$$

$$V_1 = \sqrt{2} V_{1rms} = \sqrt{2} V_{1rms} I_{rms} = \sqrt{2} * \frac{6}{\sqrt{2}} * 25.64 = 153.84$$

$$v_1(t) = 153.84 \cos(2\pi * 60t - 75.14^\circ) = 153.84 \sin(2\pi * 60 + 14.86^\circ) =$$

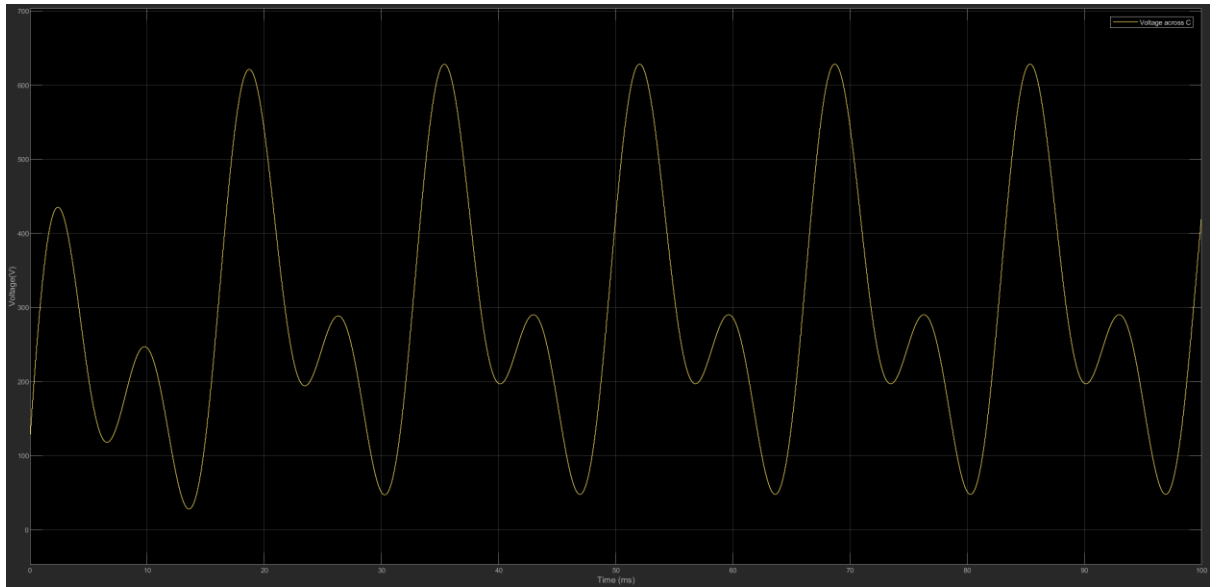


Figure 1 voltage waveform across c

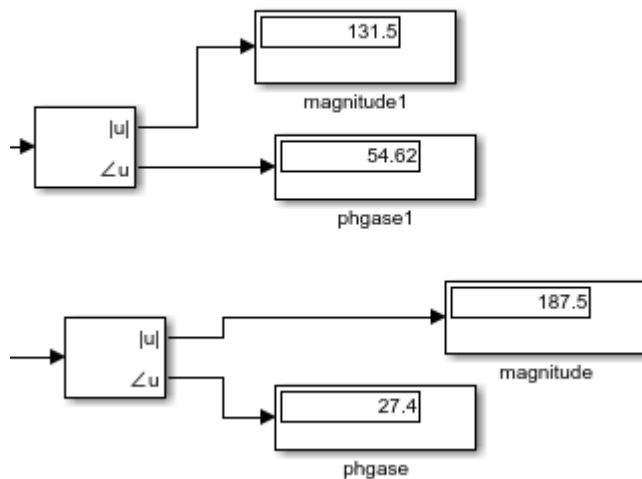


Figure 2 phase and magnitude of 1st and 2nd harmonics of voltage

Using the Fourier block we are able to find the magnitude and phase of each of the harmonics. We see the value measured are quite similar to the calculated values. Also see the magnitude of the voltage harmonics decrease as the harmonics increase.

$$\text{ii. } P_{DC} = \frac{(V_0)^2}{R} = \frac{300^2}{100} = 900W$$

$$P_{1rms} = \frac{(V_{1rms})^2}{R} = \frac{\left(\frac{187.44}{\sqrt{2}}\right)^2}{100} = 175.67W$$

$$P_{2rms} = \frac{(V_{2rms})^2}{R} = \frac{\left(\frac{153.84}{\sqrt{2}}\right)^2}{100} = 118.33W$$

$$P_{total} = P_{DC} + P_1 + P_2 = 900 + 175.67 + 118.33 = 1193.97W$$

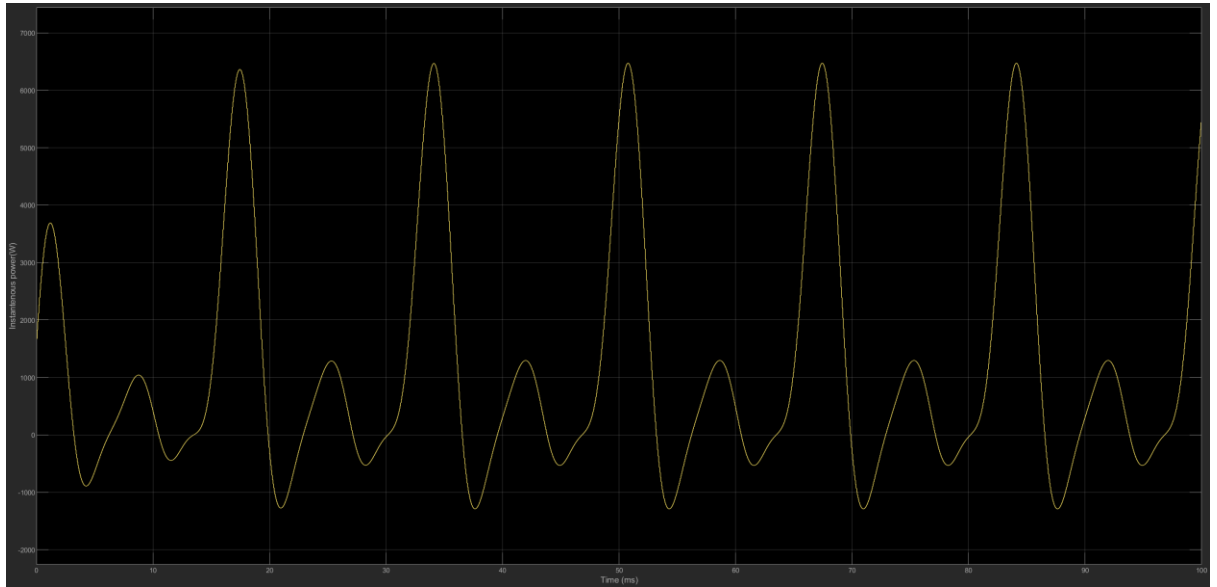


Figure 3 plot of instantaneous voltage

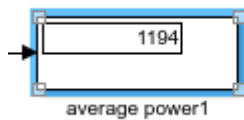


Figure 4 average power from simulation

By taking the product of the instantaneous voltage and current and then taking the average we can find the average power. We see that the measured values agree with the calculated values of 1193.97W.

By increasing the capacitance, we see a decrease in the total average power however average voltage will not decrease. While increasing the capacitance the average power of the DC component stays constant however the average power for each of the AC harmonics decreases. As the capacitance will lower the capacitors impedance at 60 and 120HZ

iii. RMS voltage

$$V_{DC} = 300V$$

$$V_{1,rms} = \frac{187.44}{\sqrt{2}} = 132.54V$$

$$V_{2,rms} = \frac{153.84}{\sqrt{2}} = 108.78V$$

$$V_{rms} = \sqrt{V_{DC}^2 + V_{1,rms}^2 + V_{2,rms}^2} = \sqrt{300^2 + 132.54^2 + 108.78^2} = 345.5V$$

RMS current

$$I_{DC} = 3A$$

$$I_{1,rms} = \frac{4}{\sqrt{2}} = 2.828A$$

$$I_{2,rms} = \frac{6}{\sqrt{2}} = 4.242A$$

$$I_{rms} = \sqrt{I_{DC}^2 + I_{1,rms}^2 + I_{2,rms}^2} = \sqrt{3^2 + 2.828^2 + 4.242^2} = 5.92A$$

To find the RMS values we are able to use a square block, clock block (to get T), integrator block and square root block and applying the following equations.

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$

$$I_{rms} = \sqrt{\frac{1}{T} \int_0^T i^2(t) dt}$$

Alternatively MATLAB also offers an RMS block which can calculate the RMS value for you.

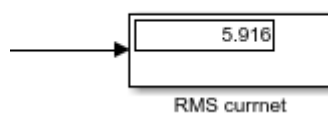


Figure 5 RMS current form simulation

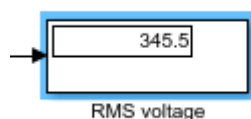


Figure 6 RMS voltage form simulation

We can see that the measured RMS values seen in figure 6 and 7 are in line with the calculated values for MRS voltage and current.

In order to find the power factor, we will make use of the following formula:

$$PF = \frac{P}{V_{rms} I_{rms}}$$

$$PF = \frac{1194}{345.5 * 5.92} = 0.584$$

To find the PF in simulation we take the RMS volt and current and input them to product block which we then feed in to a divide block along with the average power

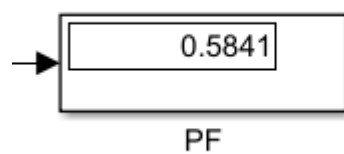
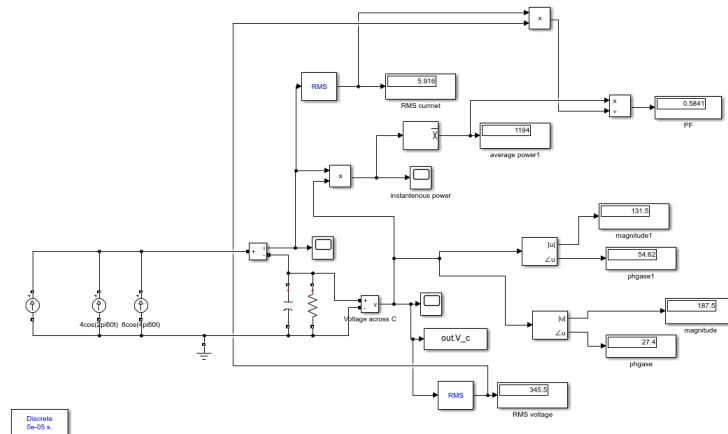


Figure 7 power factor determined in simulation

As we can see the simulated value for PF closely matches the calculated value.



Question 2

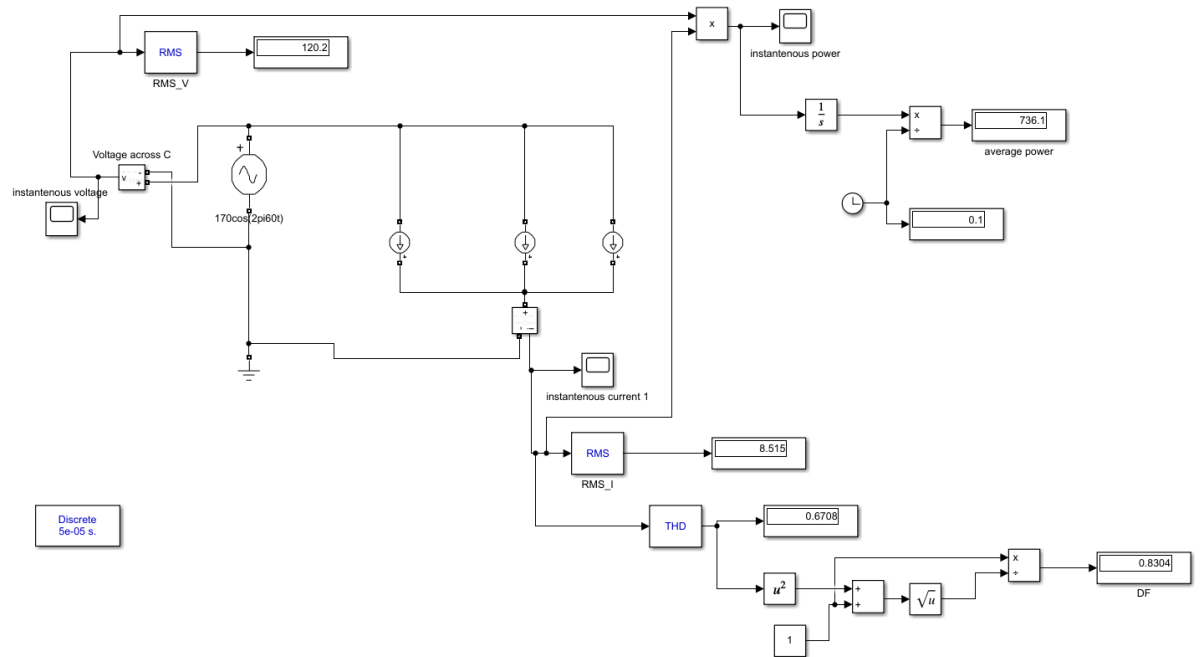


Figure 8 COMPLETE CIRCUIT FOR QUESTIONS 2

i.



Figure 9 Instantaneous voltage waveform

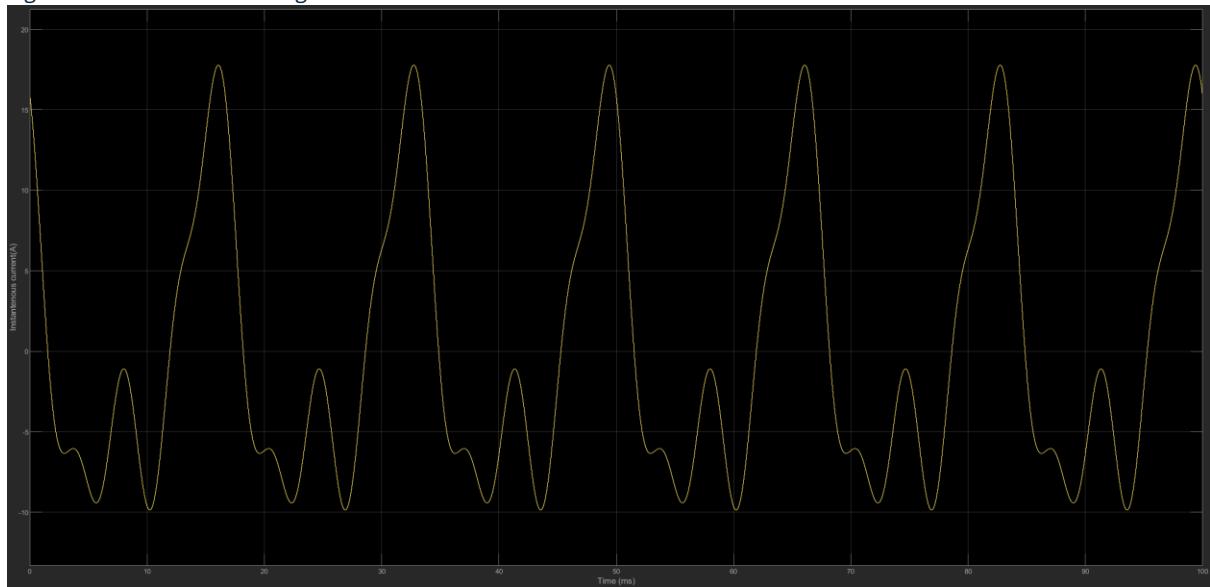


Figure 10 Instantaneous current

ii.



Figure 11 RMS current calculated using and RMS block

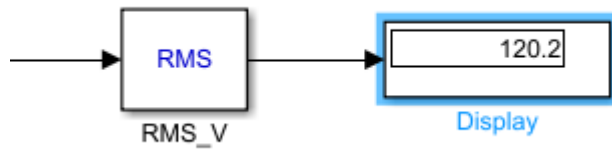


Figure 12 RMS voltage found using RMS block

iii.

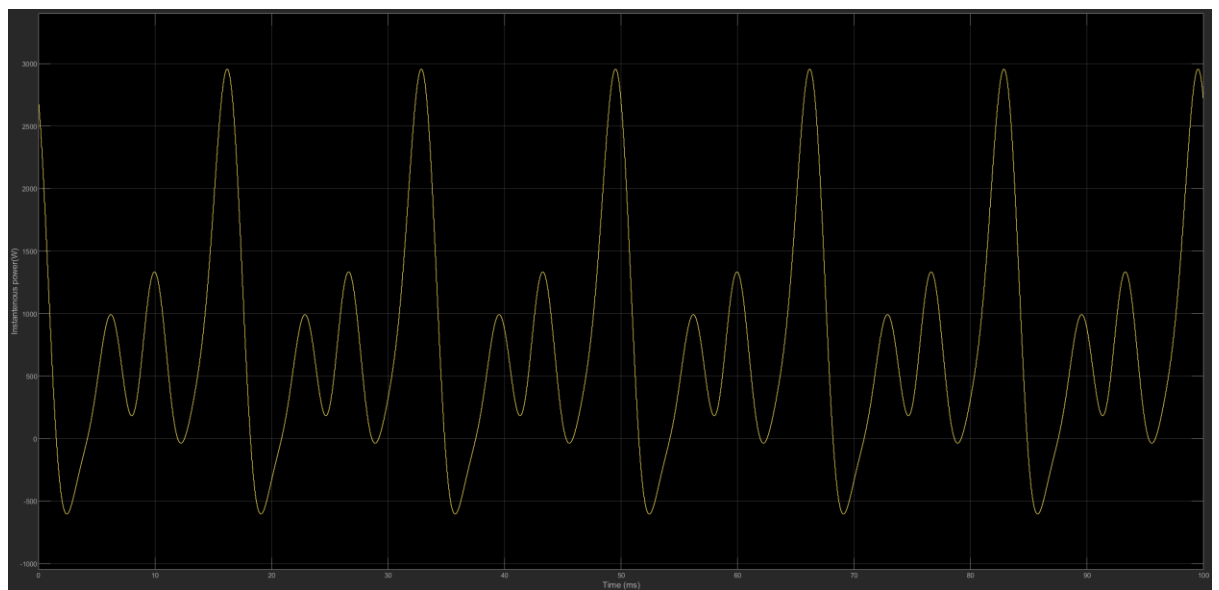


Figure 13 Instantaneous power waveform

iv.

$$P = P_{DC} + P_1 + P_2$$

$$P_{DC} = 0$$

$$P_1 = \frac{170 \times 10}{2} \cos(30^\circ) = 736.1W$$

$$P_2 = \frac{0 \times 6}{2} \cos(45^\circ) = 0W$$

$$P_3 = \frac{0 \times 3}{2} \cos(20^\circ) = 0W$$

$$P_{avg} = 736.1W$$

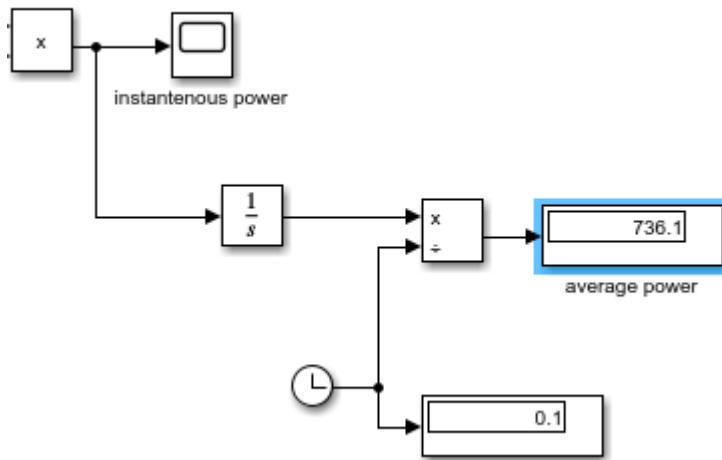


Figure 14 calculation of average power in simulation

We can see that the calculated value agree with the value form simulation.

v.

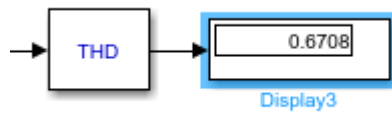


Figure 15 the THD value found using the THD block in matlab

We can then calculate the DF from the THD value

$$DF = \frac{1}{\sqrt{1 + THD^2}}$$

The above equation was simulated in MATLAB as follows.

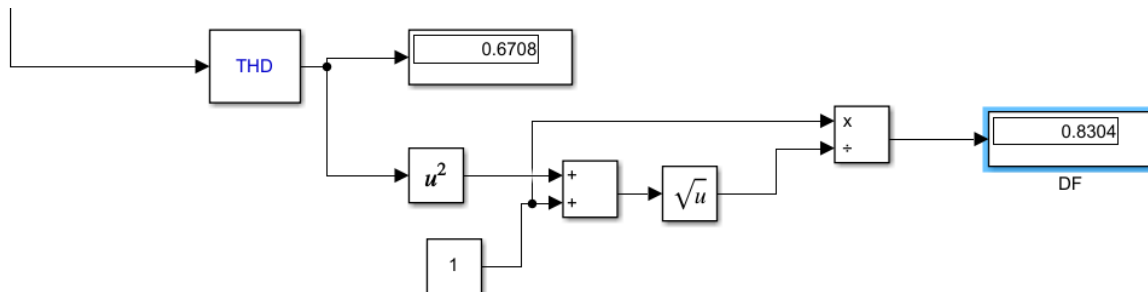


Figure 16 DF value calculation in Simulink

Ideally, we would want our DF value to be equal to 1, and have a minimum reduction in our PF. For the same reason we would want the THD value ideally equal to 0. The

measured values are not equal to the ideal values which is likely due to harmonics of the current load.

vi.

For a practical circuit we can make mention of two methods to remove unwanted harmonic (or in other words unwanted frequency components).

Method 1: Passive filters made from resistors, inductors and capacitors can remove specific harmonic components. They are typically connected in parallel with the non-linear load. At our fundamental frequency, the filter has high impedance and does not interfere with normal operation. However, at specific harmonic frequencies we wish to remove the filter has a very low impedance which essentially draws away the harmonic current to ground.

Method 2: Active Power Filters Active power filters are a type of power electronics device. This device can cancel out harmonic components in real time by injecting compensating currents. They measure the load current then use a controller to calculate the harmonic content, then they generate equal but opposite harmonic current to cancel out these components.

vii. In part (i) it is observed that you apply a sinusoidal voltage waveform across a nonlinear load, you develop a current through the load whose waveform is non-sinusoidal. This sinusoidal waveform contains harmonics and can therefore be represented by a Fourier series. In (iv) we see when trying to determine the average power delivered to the load, that the only non-zero power term is at the frequency of the applied voltage. This means there is additional power contained in other harmonics that is not being absorbed by the load. Therefore, the harmonics lead to a distortion in the power factor which can be represented by the DF (distortion factor) which we calculated in (v).

$$PF = \cos(\theta_1 - \phi_1) \times DF$$

vii. According to international standards such as the IEEE 519, the maximum allowable THD limit on voltage from the grid is 5%. Voltage THD can be seen as how non-linear loads distort the grid voltage. Increasing current THD can contribute to this voltage distortion, which negatively affect the grid by reducing its stability and power transfer quality. This can also negatively affect other devices connected to the grid, hence why the THD is limited. Therefore, it's very important for loads connected to the grid to be managed as to not exceedingly increase the grid voltage THD

